



Unity V3

Technical White Paper

Table of Contents

1.	Introduction	3
2.	Evolution of Optical Technology: From Mechanical Precision to AI	3
	Innovations Transforming Vision	3
	Intelligent Optics: Innovation and Future	3
3.	Artificial Intelligence: The New Engine of Optical Design	4
	Algorithm Training	5
	Personalization Based on Refractive Status	6
	Personalization Based on the Cylinder	8
	From Data to Decision: Data-Driven Philosophy and System Training	8
4.	InnoVue Extreme Technology: Understanding the Impact of Visual Precision	9
	Unity V3: Redefining Progressive Lens Intelligence	10
	From Clinical Data to Functional Design	10
	A New Metric: High-Impact Astigmatism	11
5.	Unique Value Proposition	12
6.	Conclusion	13
7.	References	13

1. Introduction

Progressive lens designs have come a long way in recent years, driven by optical advances, new user expectations, and the need for more precise personalization. Until now, technological limitations have prevented us from accurately capturing how people truly see, move, and experience the world through their lenses.

The maturity of artificial intelligence (AI), virtual reality (VR), and the ability to process large volumes of data has reached a tipping point. This technological convergence allows, for the first time, a quantum leap forward in progressive lens design. Unity V3[®] redefines the standards of lens personalization, performance, and visual experience.

Thanks to the evolution of VR devices, it's now possible to access high-quality visual data that reflects how the eye behaves in different contexts. Decades of innovation have proven VR's value in clinical diagnostics, enabling practitioners to replicate real-world conditions in controlled environments with unmatched precision. What was once a simulation based on assumptions is now a fluid, credible measurement without physical constraints.

Although Unity V3 doesn't directly measure gaze dynamics, it leverages valuable information—from previous studies and gaze dynamics maps—to identify patterns, trends, and particularities. This data, combined with AI algorithms trained on thousands of real prescriptions, enables Unity to create unique lens designs that adapt not only to the user's profile but also to their characteristic visual behavior.

2. Evolution of Optical Technology: From Mechanical Precision to AI

In recent years, technology has taken an exponential leap forward, driven by digitalization, AI, and the ability to analyze large volumes of visual data. This new landscape made it

possible to develop Unity V3, which synthesizes decades of applied innovation. It's not just an incremental improvement, but a new way of understanding optical design—based on real-world data, powered by intelligent algorithms, and focused on delivering a more precise and personalized visual experience.

Innovations Transforming Vision

In an increasingly digital and interconnected world, optical innovations redefine how people interact with their surroundings. The integration of AI into visual processing is opening new doors, from accurate medical diagnostics to immersive experiences in entertainment and education. Each innovation creates a window into a future where vision goes beyond the physical, connecting us more deeply with the world around us¹.

Intelligent Optics: Innovation and Future

Recent advances in optical instrumentation have transformed the way research and vision care are approached across disciplines. The integration of AI into optical image analysis has revolutionized data interpretation, enabling faster and more informed decision-making.

The future of optics is evolving toward more personalized, connected, and user-centered experiences. Lenses that adapt automatically, visual interfaces that respond to the environment, and solutions that integrate health, style, and functionality are shaping the next generation of smart optics. Unity V3, for example, incorporates AI at the core of its architecture, transforming the development of progressive lenses. The following sections will explore how AI is applied in optical design and how this foundation enabled a breakthrough innovation that sets a new industry standard.

3. Artificial Intelligence: The New Engine of Optical Design

Historically, lens designers focused on improving material quality, geometric precision, and manufacturing efficiency. This brought major advances in progressive lenses, anti-reflective coatings, and ergonomic lens solutions. However, the approach remained static; lenses were designed for a typical profile, without considering the user's visual behavior.

Modern optics are no longer limited to correcting vision. Augmented VR devices can overlay digital information onto the physical world to enhance spatial perception, navigation, and interaction with the environment.

Thanks to the integration of VR, a milestone has been reached in the study of gaze dynamics. Gaze dynamics is a key data source for progressive lens design. Instead of adapting the user to the lens, the lens now adapts to the user.

Unity V3 leverages this capability to understand user behavior and preferences, adapting the lens design accordingly. AI made this breakthrough possible and has become a transformative tool across multiple industries—optics is no exception. In the development of Unity V3, AI is not just a supporting feature; it's the core of the design process (see Figure 1). Its ability to analyze large volumes of data and detect complex patterns enables unprecedented visual personalization, tailoring each lens to the specific needs of each user.

AI algorithms identify relationships between a user's intrinsic characteristics and the attributes of their gaze dynamics.



Figure 1: Customization of the resulting design thanks to AI

Algorithm Training

AI algorithms are a set of rules and processes that machines use to perform tasks that typically require human intelligence. These algorithms leverage large amounts of data and computation to learn, make decisions, and solve problems with minimal human intervention².

The evolution of technologies such as cloud computing, advanced processing, and big data has made artificial intelligence more accessible and efficient. There are several subgroups within AI algorithms that have different applications and techniques. The most prominent include:

1. **Machine Learning:** A core approach that enables machines to learn from data and improve over time without being explicitly programmed.
2. **Neural Networks:** Inspired by the structure of the human brain, the networks consist of multiple layers of nodes (neurons) that process data sequentially.
3. **Deep Learning:** A subcategory of machine learning that uses artificial neural networks with multiple layers (deep neural networks). These algorithms can process large volumes of unstructured data—such as images and text—and learn autonomously.
4. **Natural Language Processing (NLP):** Allows machines to interpret and understand human language. This is achieved through techniques such as syntactic and semantic analysis, which break down and examine text to understand its meaning and context.

In the Unity V3 ecosystem, AI is not an abstract concept, rather it's a concrete and applied tool. Out of the AI applications and techniques above, Unity V3 relies on machine learning.

Unlike deep learning—which requires large volumes of unstructured data and intensive computational infrastructure—machine learning works with structured and highly qualified datasets, such as optical prescriptions, vision maps, and visual usage patterns (see Figure 2).

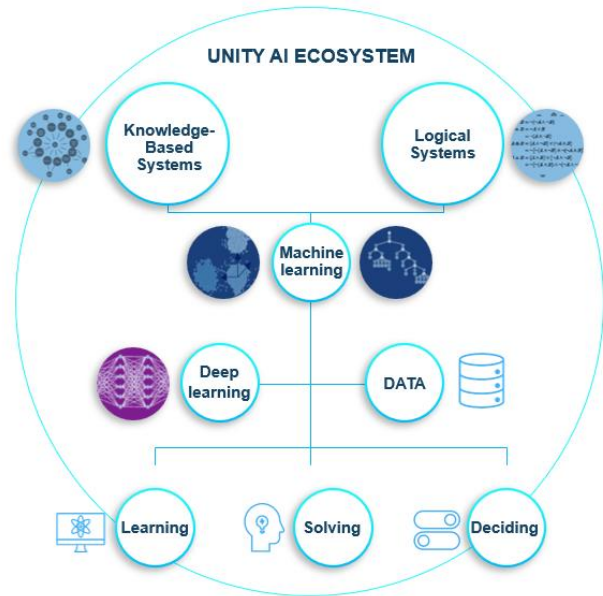


Figure 2: Unity AI ecosystem.

This allows for greater traceability, control, and efficiency in training the system without compromising design accuracy.

Unity V3 is leading the way in optical innovation by using machine learning to examine and connect multiple variables such as the power, cylinder, axis, and gaze dynamics—with users' real-world preferences. (Note that gaze dynamics refers to the mechanism that ensures that the image will be as static as possible on the retina at all times.) This data-driven strategy makes the following possible:

- **Detection of visual usage patterns across distance, intermediate, and near visual planes** is achieved through the analysis of thousands of gaze dynamics maps (VMAPs) and real-world prescription data. This approach not only identifies how users interact with different visual zones but also introduces a key differentiator: the ability to replicate and interpret these patterns in a controlled, repeatable environment.
- **Identification of underutilized lens zones** and redistribution of unwanted lateral astigmatism toward areas with low ergonomic impact.

Driven by real-world data and visual behavior insights, this lens goes far beyond technical precision. It boosts user tolerance, accelerates adaptation, and redefines what it means to see with true comfort. The result is a lens that thinks, adapts, and responds more human, more intelligent, and perfectly aligned with the demands of today's visual lifestyle.

A NEW METRIC: PERSONALIZATION BASED ON REFRACTIVE STATUS

The evolution of Unity V3 is driven by the integration of AI and a new metric that precisely correlates optical prescription data (RX), cylinder (CYL), and real visual behavior captured through gaze dynamics maps (VMAP). This metric represents a key advancement in personalized optical design, as it enables quantification of the impact that design decisions have on user experience.

By combining various optometric parameters with the population's visual behavior, we uncovered distinct design preferences (see Figure 3). This insightful analysis revealed how different factors influence people's choices and needs, enabling the delivery of more personalized and effective solutions.

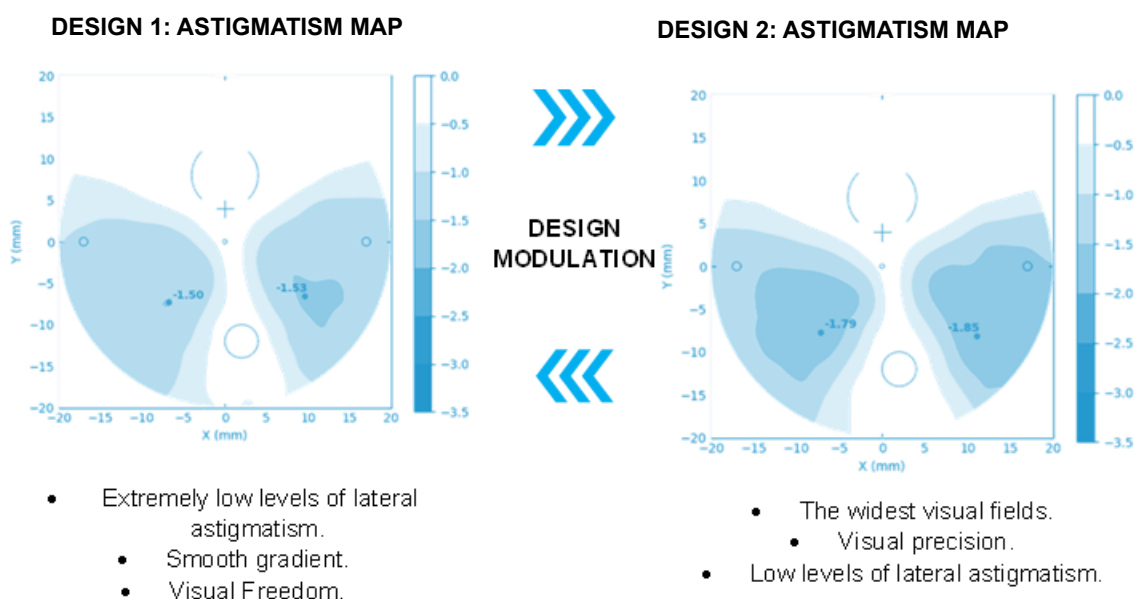


Figure 3: Design modulation.

Personalization Based on Refractive Status

In the usability test results³, we found a design preference indication based on the user's prescription. Building on this insight, we studied VMAPs (visual behavior maps) in greater depth by filtering them according to prescriptions.

Unity V3 adapts the lens design based on these preferences, optimizing the most frequently used visual zones for each group. To achieve this, we defined two main user groups according to their refractive status and prescription.

Specifically, Unity V3 considers:

1. **User's refractive status:**
 - a. Myopes
 - b. Hyperopes
2. **Prescription:**
 - a. Standard: From 0 up to +/- 5.
 - b. Extreme: Above +/- 5.

To study the correlations between prescription and users' visual preferences, **Unity V3** analyzes a key parameter: the **Monocular Gaze Angle (MGA)** (see Figure 4). This indicator reflects how each eye interacts with different zones of the lens during real-world use.

The most frequent MGA, the gaze angle that occurs with the highest intensity, helps identify characteristic visual patterns based on the user's refractive status (myopia, hyperopia, etc.) and specific prescription.

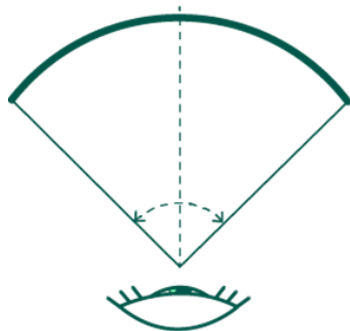
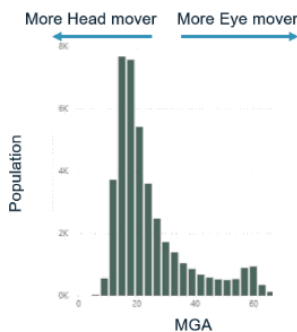


Figure 4: Representation of MGA (Monocular Gaze Angle).



Histogram representing MGA of a population

These patterns are represented using histograms (see Figure 5) that visualize the angular distribution of gaze and detect high-use zones and underutilized areas. The mode and shape of the histogram are key elements for identifying trends and establishing correlations between gaze dynamics maps and optometric parameters. While the mode indicates the most frequent gaze angle, the overall shape of the histogram determines visual usage patterns that vary by refractive status. This analysis is essential for customizing optical design because the lens geometry can be adapted to the actual visual habits of each user group, as well as for adjusting key design parameters such as lateral astigmatism distribution, power progression, and gradient smoothness.

The result is a more precise, ergonomic lens aligned with the user's real needs, especially in high visual demand situations. Below is the result of the study conducted with a total of 90,000 users analyzing 270,000 VMAPs where:

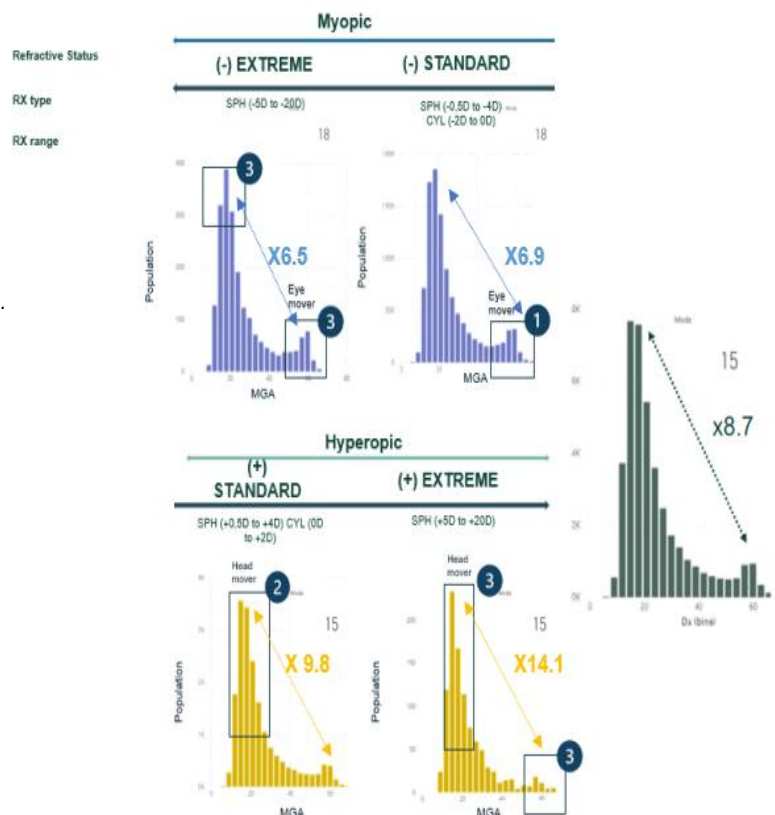


Figure 5: Histogram visualization

- 1 Myopic users exhibit greater eye movement amplitude compared to the general population.
- 2 Hyperopic users exhibit reduced eye movement amplitude compared to the general population.
- 3 The more extreme the prescription, the more pronounced this behavior is.

These gaze dynamics trends are incorporated into the progressive lens calculation to personalize the final design based on the user's refractive status and prescription.

Personalization Based on Cylinder

For years, cylinder has been treated as a simple numerical value within optical prescriptions. However, its impact goes far beyond that. In Unity V3, the cylinder value, especially the residual cylinder (the difference between the target and prescribed values) has a direct effect on user tolerance⁴.

As the cylinder value increases, even a small error in axis measurement can result in residual cylinder. This has been observed to influence preferences regarding the final design. Since most subjective cylinder measurement devices have a precision of 5°, residual cylinder is likely to occur starting at 2–3 diopters (see Figure 6).

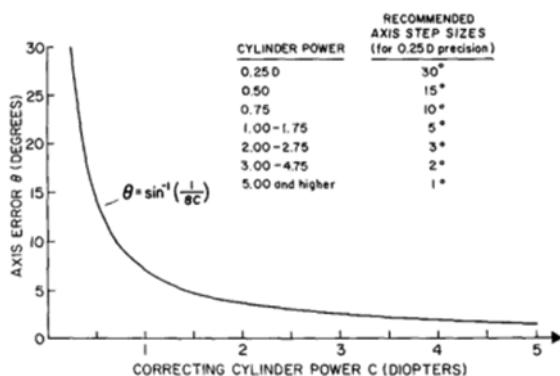


Figure 6: Residual Cylinder Representation.

At low cylinder values, residual errors are common due to low sensitivity to inaccuracies. Cylinders of 0.25D are not typically prescribed. Some individuals don't perceive differences even at values up to 0.50D, which can lead to larger prescription errors.

The combined analysis of gaze dynamics maps and knowledge of visual behavior associated with different cylinder values make it possible to identify key correlations for customizing the Unity V3 optical design.

We observed that users with low cylinder values (below 2D) show greater tolerance to moderate levels of aberrations, allowing for more design flexibility. In contrast, users with high cylinder values (above 2D) tend to be more sensitive to aberrations in their visual field, requiring highly precise optimization. These trends are integrated into the lens calculation, enabling geometry and power distribution to be tailored to individual visual profile needs. As a result, Unity V3 delivers a more personalized, ergonomic, and effective visual experience.

From Data to Decision: Data-Driven Philosophy and System Training

Unity V3 represents a breakthrough in progressive lens design by integrating AI algorithms that analyze hundreds of thousands of gaze dynamics maps alongside prescription, cylinder, and other optical parameters. This makes it possible to identify visual patterns and usage preferences that vary according to the user's refractive status (myopia or hyperopia) and the magnitude of the cylinder.

What sets Unity V3 apart is that it's the first lens on the market to combine AI with gaze dynamics data, delivering a visual solution that goes beyond traditional optical correction. To date, Unity V3 is the only lens capable of interpreting how we observe the world around us and using that understanding to create a more precise, ergonomic, and personalized visual experience. This innovation positions Unity as a leader in

intelligent optics, paving the way for a new era of personalization based on real visual behavior.

Unity V3 detects significant differences in visual behavior: myopic users exhibit greater eye movement amplitude, while hyperopic users, especially those with extreme prescriptions, show reduced eye movement amplitude. These trends are used to adapt the lens geometry to each visual profile.

Additionally, Unity V3 analyzes the cylinder value to redistribute unwanted lateral astigmatism and to precisely adjust the visual fields of the progressive lens to improve tolerance and reduce adaptation time.

Unity V3 optimizes visual precision while increasing user tolerance, significantly shortening the adaptation period. The ability to tailor optical design based on real, user-specific data marks a turning point in the evolution of progressive lenses—offering a more comfortable, effective, and individually aligned visual experience.

4. InnoVue Extreme Technology: Understanding the Impact of Visual Precision

Progressive addition lenses (PALs) are designed to provide a seamless transition in optical power from distance to near vision. This is achieved through a continuous change in surface curvature across the lens. However, this very feature introduces a fundamental optical limitation: unwanted or surface-induced astigmatism, which is an unavoidable byproduct of progressive surface geometry.

According to Minkwitz's Theorem, the rate of change of surface astigmatism is directly proportional to the rate of power change along the progression corridor. Mathematically, this is expressed as:

$$dy/dC=2 \cdot dy/dP$$

Where:

- dC/dy is the rate of change of cylinder (astigmatism).
- dP/dy is the rate of change of power along the vertical meridian.

This implies that shorter progression lengths and higher Add Powers result in steeper gradients and thus greater induced astigmatism. The induced astigmatism is not uniformly distributed. It is typically concentrated in the lateral blending zones, flanking the central progressive corridor. These areas are often referred to as the “swim zones” due to the visual artifacts they produce. The central corridor remains relatively free of astigmatism to preserve clear vision during gaze transitions. From a user's perspective, this astigmatism manifests as:

- **Blurred peripheral vision**
- **Image distortion or warping**
- **“Swim” effect** during head or eye movements

These effects are particularly noticeable during dynamic tasks such as walking, driving, or shifting gaze between objects at different distances.

Modern lens design leverages **digital freeform surfacing**, allowing for point-by-point control of lens curvature. Despite these advancements, the **optical trade-off between power progression and induced astigmatism remains a fundamental constraint** in PAL design. The challenge lies in optimizing this balance to deliver both visual clarity and ergonomic comfort.

Unity V3: Redefining Progressive Lens Intelligence

The Unity V3 algorithm was developed using multiple data sources that range from full optical prescriptions (power, cylinder, axis) to gaze dynamics captured in virtual reality environments.

This wealth of information made it possible to train an intelligent system capable of making complex decisions such as adjusting lens geometry based on the actual use of each visual zone.

As part of this process, more than 3,000 visual maps were analyzed to evaluate the impact of lens design on specific areas, particularly below the Near Reference Point (NRP). This technical validation confirmed that the benefits perceived by users far outweigh any potential visual impact, reinforcing design decisions with a solid scientific and quantifiable foundation.

It's the only lens on the market that applies artificial intelligence trained with gaze dynamics data to deliver a truly adaptive and personalized visual solution.

By redistributing unwanted lateral astigmatism toward low ergonomic impact zones, it redefines what's possible in progressive lens design—merging data science, visual behavior, and optical precision into a single, intelligent system.

From Clinical Data to Functional Design

InnoVue Extreme Technology is designed to deliver even greater visual precision by further minimizing levels of astigmatism and swim effect, ensuring enhanced peripheral vision like never before (see Figure 7).

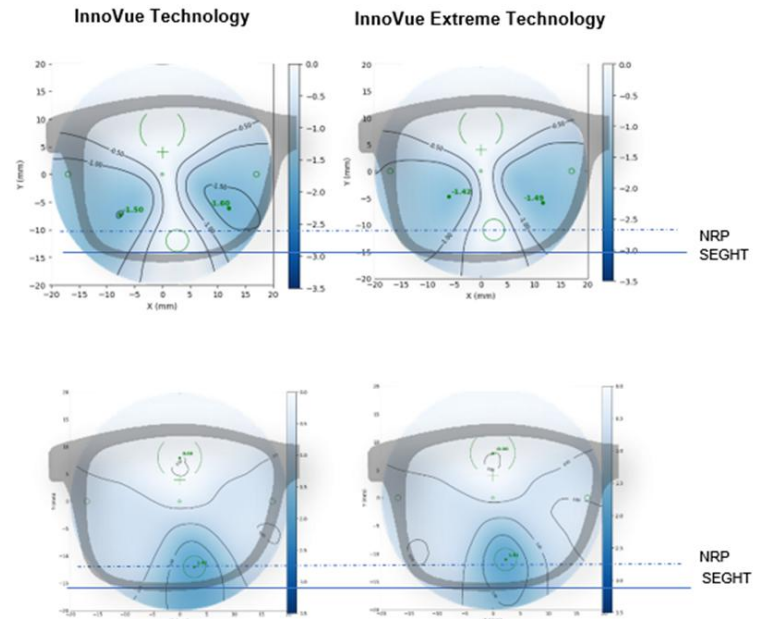


Figure 7: InnoVue vs InnoVue Extreme Technology

InnoVue Extreme technology positions the astigmatism isolines within the frame and below the near reading area in a non-ergonomic and unused zone. The amount of lateral astigmatism can be significantly improved, especially when the fitting height is greater than the corridor length.

1. Users experience a noticeable improvement in visual quality, as the total astigmatism within the frame contour is significantly reduced. This is supported by the results of an internal user test conducted with 22 participants, which confirmed that 85% of users are very satisfied with the InnoVue Extreme Technology³.
2. 3 out of 4 users prefer InnoVue Extreme technology over the standard InnoVue Technology³.

A New Metric: High-Impact Astigmatism

By softening the most critical astigmatism zones, InnoVue Extreme allows users with demanding prescriptions to enjoy a more stable, natural, and comfortable visual experience.

Reducing lateral astigmatism improves the optical design and redefines the user experience, delivering a sense of fluidity and clarity with every glance.

Thanks to the significant reduction in lateral astigmatism (see Figure 8), what was once considered a high-intensity zone is now classified as medium-intensity astigmatism. This improvement directly impacts the quality of the design and the user's comfort.

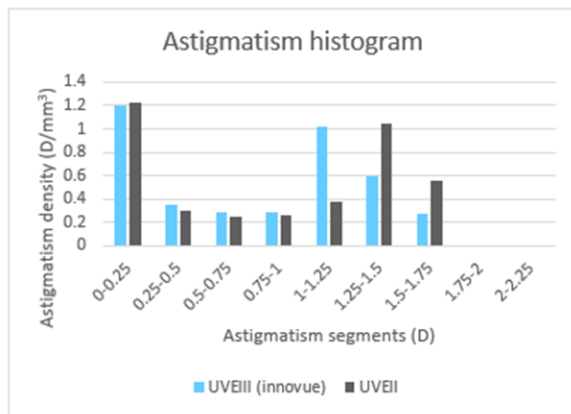


Figure 8: Minimization of High-Impact Astigmatism.

Since low astigmatism* up to 1 diopter (1D) minimal impact on the user's visual acuity, it's essential that progressive lens designs closely analyze unwanted astigmatism in the lens periphery above that threshold.

**Astigmatism is measured in diopters, which allow it to be classified as Low, Medium, or High. Low astigmatism ranges from 0.25D to 1D, medium from 1D to 3D, and high is considered anything above 3D.*

This peripheral astigmatism significantly affects the end user's comfort, adaptation, and visual

perception, and its control and distribution are used to evaluate lens performance.

Comparisons between InnoVue and InnoVue Extreme show substantial reduction in these critical zones, improving optical performance and user experience (Figure 9).

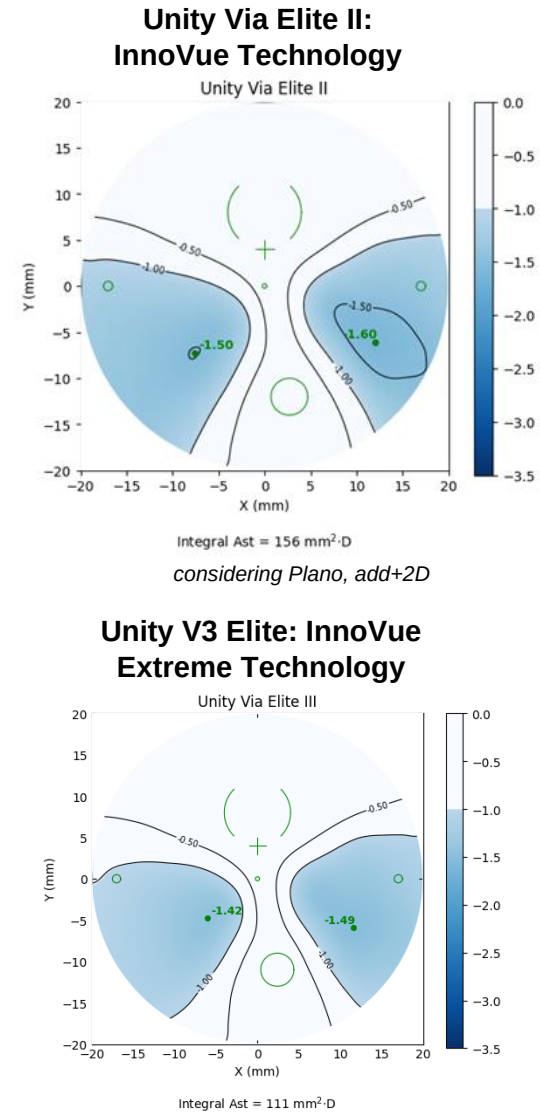


Figure 9: UNITY VIA ELITE II vs UNITY V3

To quantify High-Intensity Astigmatism (HIA) in each design, the values of astigmatism greater than 1 diopter (1D) are summed within a 20 mm radius circle centered on the optical center of the design (our standard area of interest), see Figure 10.

SPH 0D, add +2D	Maximum lateral astigmatism (D)	HIA (mm ² ·D)
UNITY VIA ELITE II	1.60D T / 1.50D N	156
UNITY V3	1.49D T / 1.42D N	111

Figure 10: Comparative Summary of HIA Between UVEII and UVEIII

Unity V3 shows a reduction in both the temporal and nasal zones compared to Unity Via Elite™ II, indicating an improvement in design smoothness. Additionally, Unity V3 reduces High-Impact Astigmatism (HIA) by **30%** compared to Unity Via Elite II (from 156 to 111 mm²·D), resulting in a more comfortable visual experience for the user.

5. Unique Value Proposition

Unity V3 represents the pinnacle of customized ophthalmic lens design, combining AI with advanced optical engineering to deliver a highly personalized visual experience.

These lenses are designed for modern visual demands and are ideal for patients seeking precision, comfort, and adaptability in their eyewear. Unity V3 integrates additional advanced technologies that work together to boost optical performance and elevate the user experience:

Benefits for the End User:

- Unity V3 lenses are tailored considering gaze dynamics trends.
- **Designed for real users:** continuous innovation through real-world data.

- **Greater visual precision and comfort:** The 30% reduction in high-impact astigmatism (HIA) improves peripheral clarity and reduces visual strain.
- **Intelligent design build considering 90,000 profiles and 270,000 gaze maps.**
- **Shorter adaptation time, ergonomics and comfort:** The ergonomic redistribution of astigmatism provides users with a smoother and more natural transition to the new design.
- **Improved tolerance and overall satisfaction:** The optimized design results in a more stable, comfortable, and satisfying visual experience from the very first use.

Advantages for Eye Care Professionals:

- **Fewer non-adapts and remakes through faster adaptation:** while all PALs require some degree of post-fitting adjustments due to their inherent optical complexity, Unity V3 significantly reduced the rate of non-adapts and remakes. Its intelligent design facilitates faster adaptation by aligning lens geometry with real gaze behavior, minimizing perceptual distortion from the outset and enhancing user satisfaction from day one.
- **Strong technical argumentation:** The use of AI and objective metrics provides clear tools to communicate the product's value.
- **Competitive differentiation:** Unity V3 enables professionals to offer a premium solution backed by real data and advanced technology.

6. Conclusion

Unity V3 replaces static, rule-based optical design with a dynamic system that learns and evolves from real-world data. Hundreds of thousands of prescriptions, gaze dynamics maps, and usage patterns have been processed through AI to create a progressive lens that better adapts to modern demands. Unlike traditional approaches, where optical design was based on static models and predefined rules, Unity V3 introduces a dynamic and revolutionary philosophy: the system learns, trains, and evolves using real-world data.

It is the first solution to translate this understanding into true progressive lens optimization—more precisely adjusting geometry and power distribution to each profile. The result is a major leap in personalized optics: a lens that not only corrects vision, but interprets, responds to, and anticipates user needs.

7. References

¹ Visual Care. (s.f.). Desarrollo tecnológico en el mundo de la óptica. Visual Care.

² KeepCoding. (s.f.). Inteligencia artificial y los algoritmos: ¿cómo se relacionan? KeepCoding.

³ Horizons Optical, Internal Studies.

⁴ Study in collaboration with Universidade do Minho in Braga, Portugal.